

Optimal control of Dynamical Systems

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Problem 1: Derivatives and Euler-Lagrange Equation (20+20 points)

1. Given $F = x\sqrt{y^2 + (y')^2}$ where $y = y(x)$:

(a) Find partial derivatives:

a.1 Find $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$, $\frac{\partial F}{\partial y'}$

Solution:

$$\begin{aligned}\frac{\partial F}{\partial x} &= \sqrt{y^2 + (y')^2} \\ \frac{\partial F}{\partial y} &= \frac{xy}{\sqrt{y^2 + (y')^2}} \\ \frac{\partial F}{\partial y'} &= \frac{xy'}{\sqrt{y^2 + (y')^2}}\end{aligned}$$

a.2 Derive $\frac{dF}{dx}$ using the above partial derivatives

Solution: By the chain rule for total derivatives:

$$\begin{aligned}\frac{dF}{dx} &= \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{d^2y}{dx^2} \\ &= \sqrt{y^2 + (y')^2} + \frac{xy}{\sqrt{y^2 + (y')^2}} \cdot y' + \frac{xy'}{\sqrt{y^2 + (y')^2}} \cdot y'' \\ &= \frac{y^2 + (y')^2 + xyy' + xy'y''}{\sqrt{y^2 + (y')^2}}\end{aligned}$$

(b) For $y = \sin x$:

b.1 Substitute into F and compute $\frac{dF}{dx}$ directly

Solution: Given $y = \sin x$, $y' = \cos x$, $y'' = -\sin x$:

$$\begin{aligned}F &= x\sqrt{\sin^2 x + \cos^2 x} = x \cdot 1 = x \\ \frac{dF}{dx} &= 1\end{aligned}$$

b.2 Verify consistency with (a) results

Verification: Substitute into part (a) result:

$$\begin{aligned}
\frac{dF}{dx} &= \frac{\sin^2 x + \cos^2 x + x \sin x \cos x + x \cos x(-\sin x)}{1} \\
&= \frac{1 + x \sin x \cos x - x \sin x \cos x}{1} \\
&= 1 \quad (\text{Consistent with direct computation})
\end{aligned}$$

4. For the functional:

$$S[y] = \int_a^b ((y')^2 + y) dx \quad \text{with} \quad y(0) = 0, \quad y(1) = 2$$

(a) **Show stationary paths satisfy : $2y'' - 1 = 0$ with $y(0) = 0, y(1) = 2$**

Solution:

First Identify the Lagrangian:

$$L(y, y') = (y')^2 + y$$

Then Compute Euler-Lagrange equation:

$$\begin{aligned}
\frac{\partial L}{\partial y} - \frac{d}{dx} \left(\frac{\partial L}{\partial y'} \right) &= 0 \\
1 - \frac{d}{dx}(2y') &= 0 \quad \Rightarrow \quad 1 - 2y'' = 0 \\
2y'' - 1 &= 0
\end{aligned}$$

Last Boundary conditions are given as:

$$y(0) = 0, \quad y(1) = 2$$

(b) **Solve the ODE to prove the stationary path is : $y(x) = \frac{1}{4}x^2 + \frac{7}{4}x$:**

Solution:

First Solve the ODE:

$$2y'' - 1 = 0 \quad \Rightarrow \quad y'' = \frac{1}{2}$$

First integration:

$$y' = \frac{1}{2}x + C_1$$

Second integration:

$$y = \frac{1}{4}x^2 + C_1x + C_2$$

Then Apply boundary conditions:

$$\begin{aligned}
y(0) &= C_2 = 0 \\
y(1) &= \frac{1}{4} + C_1 = 2 \quad \Rightarrow \quad C_1 = \frac{7}{4}
\end{aligned}$$

Last Final solution:

$$y(x) = \frac{1}{4}x^2 + \frac{7}{4}x$$

Problem 2: First Travel Problem (Fermat's Principle, 40 points)

Snell's Law Derivation

Fermat's principle states that light travels along a path that minimizes the total travel time. For a path $y(x)$ in a medium with position-dependent speed $v(x, y)$, the total travel time is given by the functional:

$$J[y] = \int_a^b \frac{\sqrt{1 + (y')^2}}{v(y)} dx$$

where the speed of light depends only on the y -coordinate (i.e., $v(x, y) = v(y)$) and is piecewise constant across the x -axis (boundary between two media):

$$v(y) = \begin{cases} c_1 & \text{for } y > 0 \quad (\text{speed in first medium}) \\ c_2 & \text{for } y < 0 \quad (\text{speed in second medium}) \end{cases}$$

1. Apply the Euler-Lagrange equation to the functional $J[y]$ to show that:

$$\frac{1}{v(y)\sqrt{1 + (y')^2}} = C$$

where C is a constant.

Solution:

The Euler-Lagrange equation for a functional $J[y] = \int_a^b F(x, y, y') dx$ is:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

Since F do not depend on x , so Euler-Lagrange equation can be simplified as:

$$F - y' \cdot \frac{\partial F}{\partial y'} = C$$

For the given functional, the integrand $F = \frac{\sqrt{1 + (y')^2}}{v(y)}$.

- Compute $\frac{\partial F}{\partial y'}$:

$$\frac{\partial F}{\partial y'} = \frac{1}{v(y)} \cdot \frac{y'}{\sqrt{1 + (y')^2}}$$

- Substitute get: $\frac{\sqrt{1 + (y')^2}}{v(y)} - y' \cdot \frac{1}{v(y)} \cdot \frac{y'}{\sqrt{1 + (y')^2}} = C$.

- Simplify gives: $\frac{1}{v(y)\sqrt{1 + (y')^2}} = C$ (where C is a constant).

2. Explain why $y'(x)$ is a piecewise constant function when $v(y)$ is piecewise constant.

Solution:

From part (1), we have $\frac{1}{v(y)\sqrt{1 + (y')^2}} = C$. Rearranging gives:

$$\sqrt{1 + (y')^2} = \frac{1}{C \cdot v(y)}$$

Squaring both sides:

$$(y')^2 = \left(\frac{1}{C \cdot v(y)} \right)^2 - 1$$

Since $v(y)$ is piecewise constant ($v(y) = c_1$ for $y > 0$ and $v(y) = c_2$ for $y < 0$), the right-hand side of the equation is constant in each medium. Thus:

For $y > 0$: $(y')^2 = \left(\frac{1}{C \cdot c_1}\right)^2 - 1$ (constant), so y' is constant.

For $y < 0$: $(y')^2 = \left(\frac{1}{C \cdot c_2}\right)^2 - 1$ (constant), so y' is constant.

Therefore, $y'(x)$ is piecewise constant.

3. Using the results from parts (1) and (2), prove Snell's Law:

$$\frac{\sin \alpha_1}{c_1} = \frac{\sin \alpha_2}{c_2}$$

where α_1, α_2 are the angles between the light ray and the boundary normal in each medium.

Solution:

From part (1), the constant C holds for both media:

$$\frac{1}{c_1 \sqrt{1 + (y'_1)^2}} = C \quad \text{and} \quad \frac{1}{c_2 \sqrt{1 + (y'_2)^2}} = C$$

where y'_1, y'_2 are the slopes of the path in each medium.

- Relate slope to angle with the boundary: Let φ_1, φ_2 be the angles between the light ray and the x -axis (boundary). Then $y' = \tan \varphi$, and using $1 + \tan^2 \varphi = \sec^2 \varphi$:

$$\sqrt{1 + (y')^2} = \sec \varphi = \frac{1}{\cos \varphi}$$

- Substitute into the equation from part (1): For $y > 0$: $\frac{\cos \varphi_1}{c_1} = C$; for $y < 0$: $\frac{\cos \varphi_2}{c_2} = C$. Equating the two expressions for C :

$$\frac{\cos \varphi_1}{c_1} = \frac{\cos \varphi_2}{c_2}$$

- Since angles satisfies: $\alpha = \frac{\pi}{2} - \varphi$, so:

$$\frac{\sin \alpha_1}{c_1} = \frac{\sin \alpha_2}{c_2}$$

which is Snell's Law.

Problem 3: Second Travel Problem (Pontryagin Principle, 50 points)

Car Control Optimization

Maximize:

$$P[\alpha] = - \int_0^T \frac{1}{2} k \alpha^2(t) dt + s(T)$$

subject to:

$$\begin{cases} \dot{s}(t) = v(t), & s(0) = 0 \\ \dot{v}(t) = l\alpha(t), & v(0) = 0 \\ \alpha(t) \in [-1, 1] & \forall t \end{cases}$$

1. Formulate Hamiltonian:

$$H(s, v, p_1, p_2, \alpha) = p_1 v + p_2 l \alpha - \frac{1}{2} k \alpha^2$$

Solution:

According to Pontryagin's Maximum Principle (PMP), the Hamiltonian combines the system dynamics and cost function. $H(x, p, \alpha) = p \cdot f(x, \alpha) + L(x, \alpha)$

The dynamics terms come from $\dot{s} = v$ and $\dot{v} = l\alpha$, contributing $p_1 v + p_2 l\alpha$.

The cost function term is $-\frac{1}{2}k \max(0, \alpha^2)$, representing fuel consumption (only when accelerating).

2. Why:

$$p_1^*(T) = 1, \quad p_2^*(T) = 0$$

Solution:

For

$$p_i(T) = \frac{\partial \Phi}{\partial x_i(T)}$$

where Φ is the terminal cost term in the objective function

The performance measure $P[\alpha]$ depends explicitly on the terminal position $s(T)$ but not on terminal velocity $v(T)$. Thus:

$$p_1^*(T) = \frac{\partial P}{\partial s(T)} = 1, \quad p_2^*(T) = \frac{\partial P}{\partial v(T)} = 0$$

3. Why:

$$p_1^*(t) = 1, \quad p_2^*(t) = T - t$$

Solution:

The costate equations are:

$$\dot{p}_1 = -\frac{\partial H}{\partial s} = 0, \quad \dot{p}_2 = -\frac{\partial H}{\partial v} = -p_1$$

Solving with terminal conditions:

For p_1^* : $p_1^* = 0$ with $p_1^*(T) = 1$ gives $p_1^*(t) \equiv 1$

For p_2^* : $p_2^* = -1$ with $p_2^*(T) = 0$ gives $p_2^*(t) = T - t$

4. Why the maximal obtains at $\alpha^*(t) = \frac{l(T-t)}{k}$

Solution:

For $a > 0$, the Hamiltonian becomes:

$$H = v + (T - t)l\alpha - \frac{1}{2}k\alpha^2$$

Taking derivative :

$$\frac{\partial H}{\partial \alpha} = l(T - t) - k\alpha = 0 \Rightarrow \alpha = \frac{l(T - t)}{k}$$

Second derivative confirms maximum: $\frac{\partial^2 H}{\partial \alpha^2} = -k < 0$

5. Why:

$$\alpha^*(t) = \min\left(\frac{l(T - t)}{k}, 1\right)$$

Solution:

Physical constraint requires $\alpha \in [-1, 1]$

Two phases:

(a) *Maximum acceleration phase*: When $\frac{l(T-t)}{k} \geq 1$ (early time)

(b) *Linear decay phase*: When $\frac{l(T-t)}{k} < 1$ (later time)

The min function enforces the control constraint

6. Draw the graph of $\alpha^*(t)$

